CS 24 AP Computer Science A Review

Week 4: Iteration

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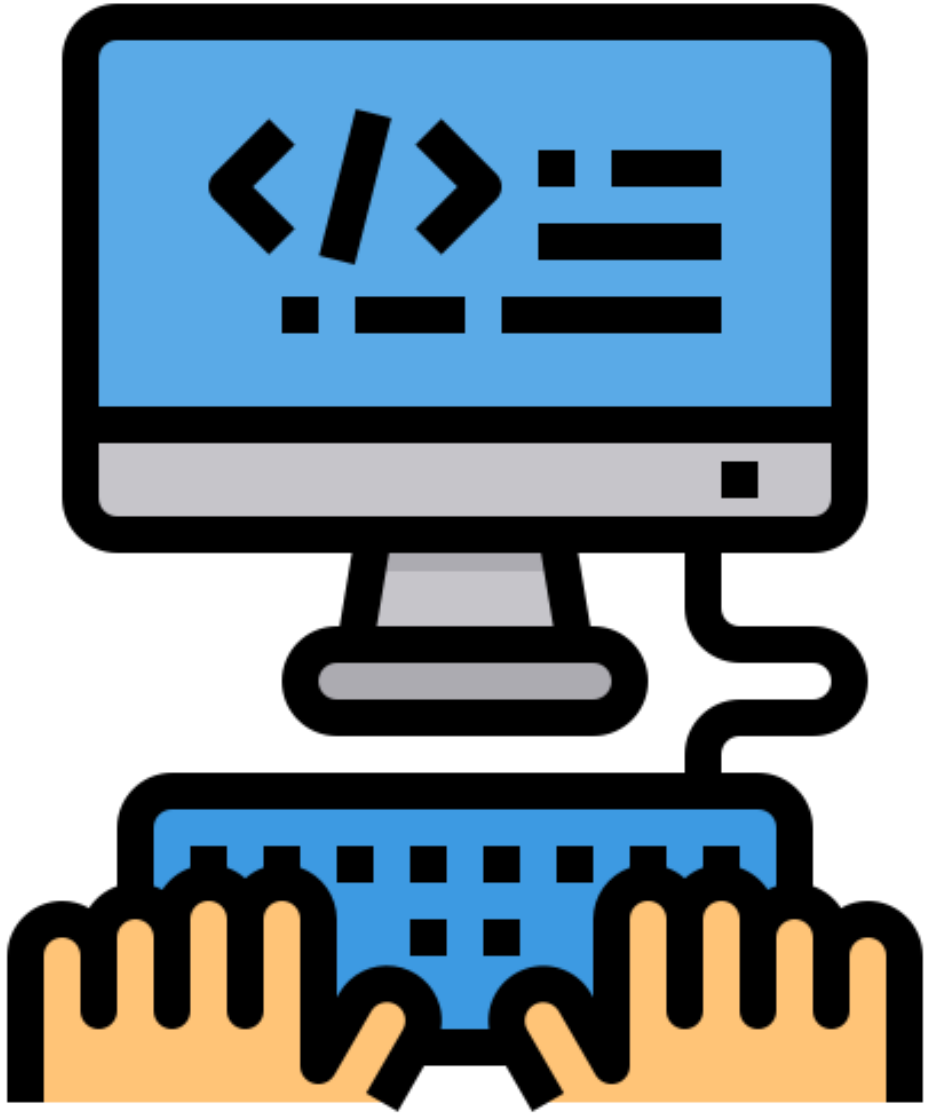
Topics

- Swap Operation
- The while statement
- The for-statement
- The for-each statement
- Multiple Choice Questions
- Loop Algorithms



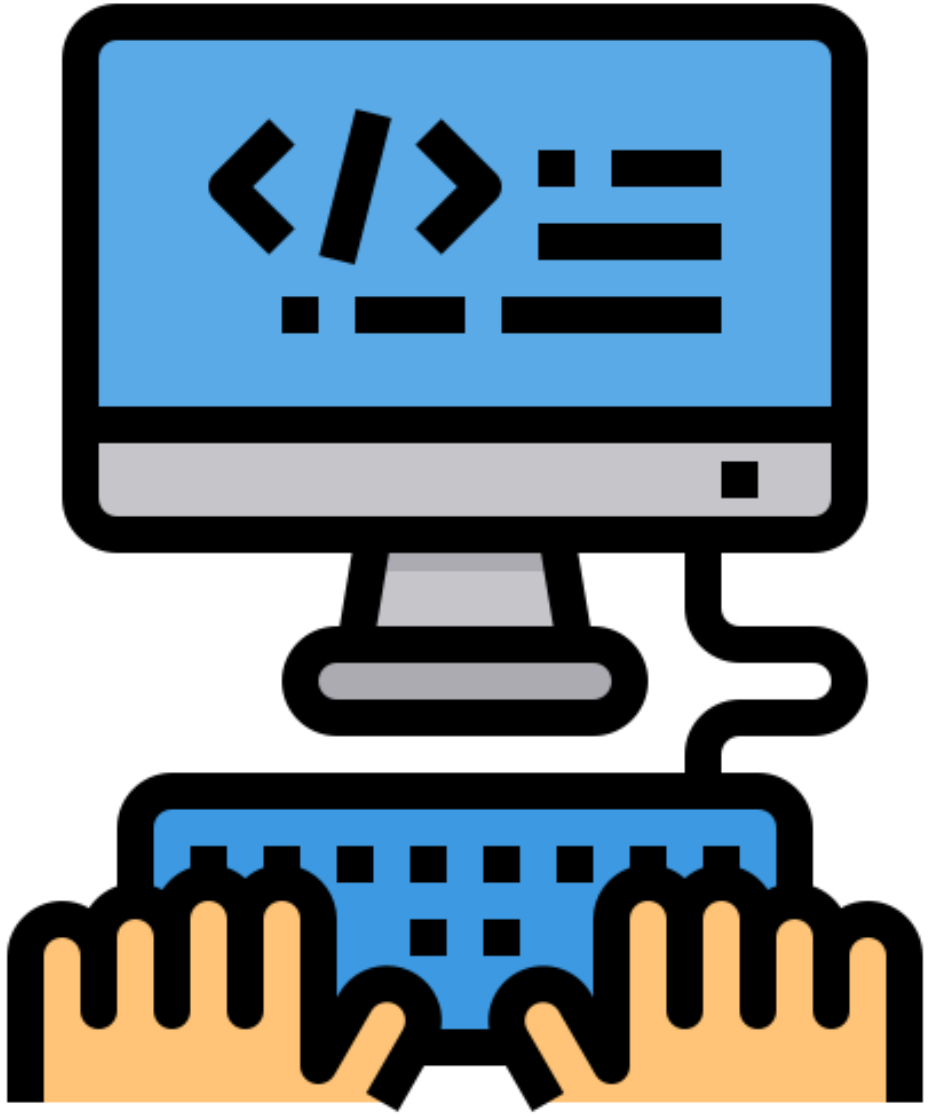
Swap Operation

SECTION 1

Swap Operation

Swap two integer a, b:

```
int tmp = a;
a = b;
b = tmp;
```



While Statement

SECTION 2

Loops for Repetition

100 times

```
System.out.println("Welcome to Java!");
System.out.println("Welcome to Java!");
System.out.println("Welcome to Java!");
System.out.println("Welcome to Java!");
System.out.println("Welcome to Java!");
System.out.println("Welcome to Java!");
…
…
…
System.out.println("Welcome to Java!");
System.out.println("Welcome to Java!");
System.out.println("Welcome to Java!");
```

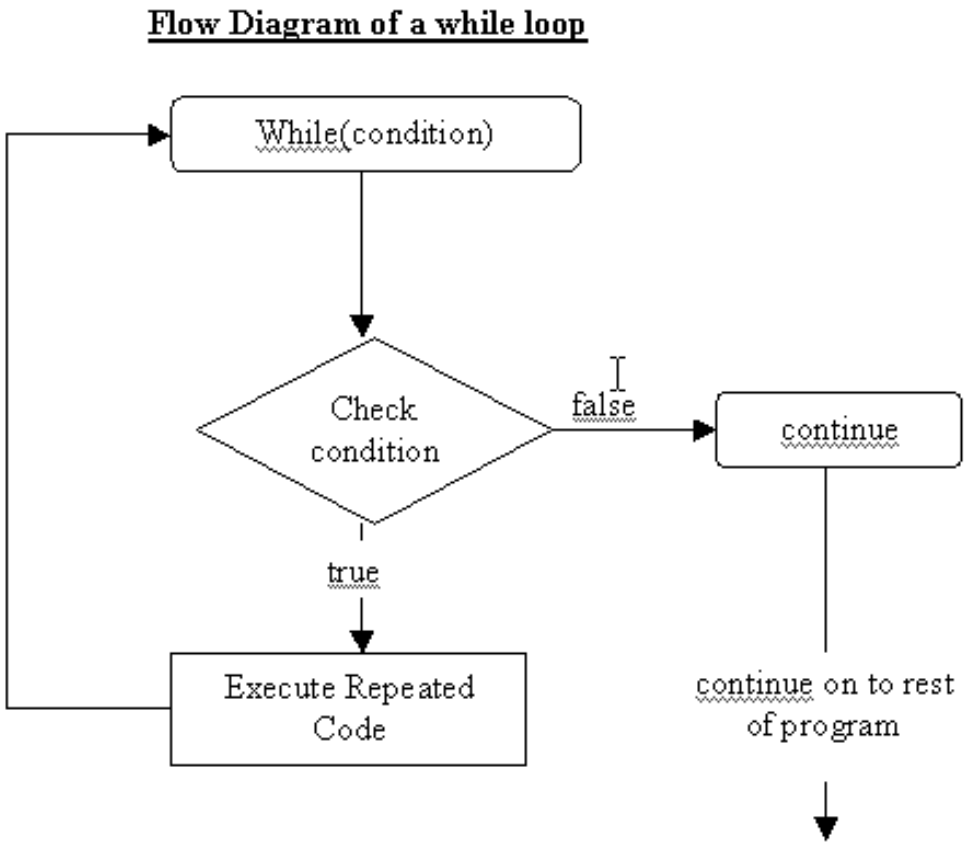
Solution to it: while-loop

```
int count = 0;
while (count < 100) {
   System.out.println("Welcome to Java");
   count++;
}
```

while loop

The syntax for the while loop is:

```
while (loop-continuation-condition)
   // loop body
   Statement(s);
}
```



Comparison of if-statement and while-loop

```
int x = 0;                    // if-statement
if (x < 10) {
    System.out.println("Welcome to Java.");
}

int x = 0;                    // while-loop
while (x < 10) {
    System.out.println("Welcome to Java.");
    x++;
}
```

While Statement

```
while (continue condition A){
    // some statements
}
// exit condition (post condition): -A, or not A
```

Equivalent For-Loop

```
int count = 0;
while (count < 100) {
   System.out.println("Welcome to Java");
   count++;
}
```

Initial Condition: int count = 0;
Continue Condition: count < 100
Update Statement: count++

Equivalent For-Loop

```
for (int count=0; count <100; count++){
  System.out.println("Welcome to Java!");
}
```

Initial Condition: int count = 0;
Continue Condition: count < 100
Update Statement: count++

Continue Condition

- Opposite of Stop Condition. If the stop condition is count >=100, then, the continue condition is count <100;
- Continue condition may have complicated logic combination
- Sentinel variable !str.equals(“exit”)
- Sentinel variable num != 999
- Prevention of array index out of bound

Exit Condition (Stop Condition)

- The opposite of continue condition.
- After the loop operation, exit condition will always be met.

Infinite Loop

A loop that will never get out.

```
int a= 0;
while (a < 5){
    System.out.print("Do something.");
}
```

Definite Loop

A loop that will get out for sure.

```
int a= 0;
while (a < 5){
    System.out.print("Do something.");
    a++;
}
```

In-deterministic Loop

A loop that may or may not get out depending on odds.

```
int a= 0;
while (a < 5){
    System.out.print("Do something.");
    a = (int)(Math.random()*10);
}
```

Multiple Choice (1)

Consider the following code segment:

```
int val1 = 2,val2 = 22, val3 = 78;
while (val2 % val1==0 ||val2%3==0){
    val3++;
    val2--;
}
```

Which will val3 contain after the code segment is executed?

Multiple Choice (1)

Ⓐ 78

Ⓑ 79

Ⓒ 80

Ⓓ None of the above

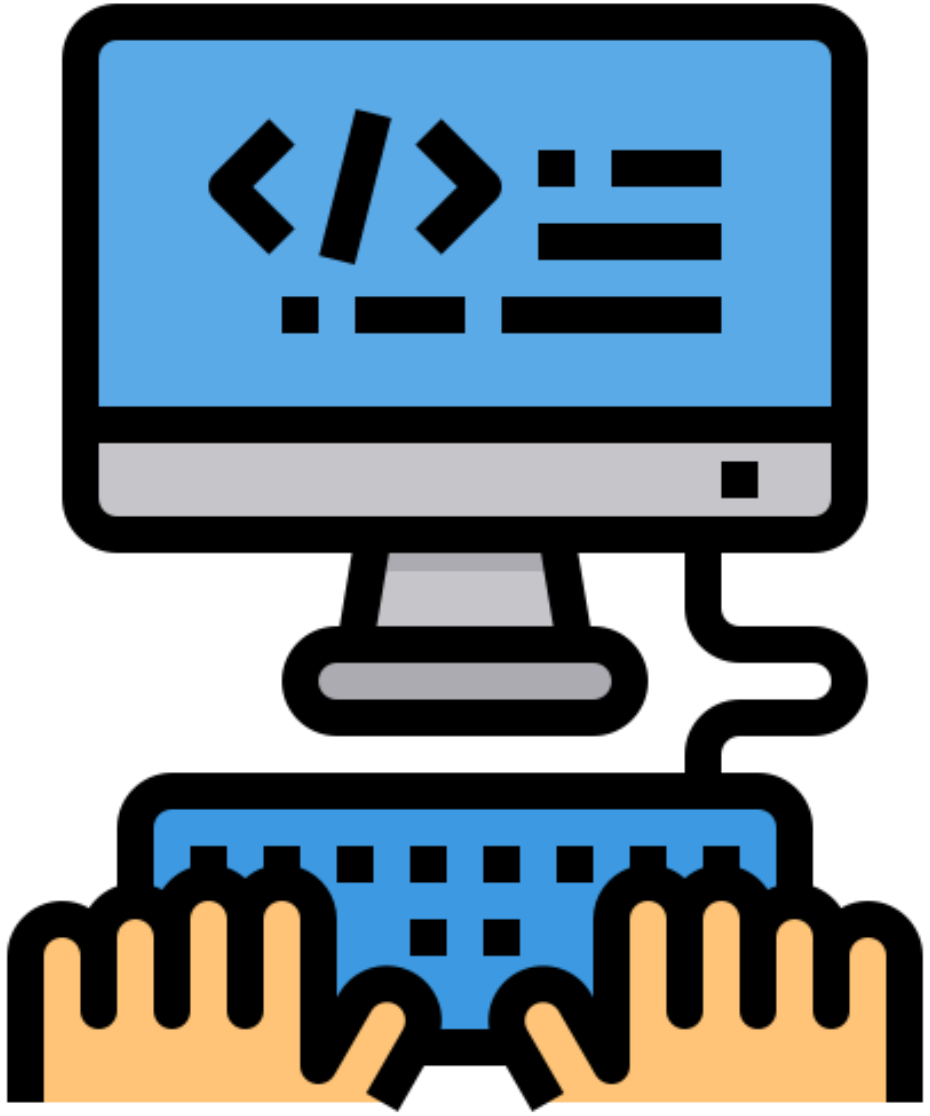
Multiple Choice (1)

Ⓐ 78

Ⓑ 79

Ⓒ 80

Ⓓ None of the above



for Statement

SECTION 3

for loop

Syntax of for loop:

```
for (initial_condition; continuation-condition; action-after-each-iteration)
{
  // loop body:
  Statement(s);
}
```

For-loop Statement

Declaring and Initializing loop control variable

Checking condition

Incrementing loop control variable

```
for (int i =0; i<10 ; i++) {

// Loop statements to be executed

}
```

For-Loop

- For loops are mostly definite loop but not always are.
- For loop can be equivalent to while loop (take out initial condition and update statement)
- Counter loop
- Index loop
- Jumping conditional loop

Initial Condition

- What is your base case?
- Loop design is similar to recursion design

Continue Condition

- What is your exit condition?
- Similar to the continue condition of while-loop (exit condition similar)

Update Operation

- What are your steps?
- for (int i=0; i<str.length()-2; i++); // window of 3
- for (int i=0; i<str.length(); i+= 2); // index jump by 2
- for (int i=0; i<str.length(); i++) if (i%4==0 || i%4==1) …; // 0, 1, 4, 5, 8. 9
- for (int i=0; i<str.length()/3*3; i+= 3){} // group of 3.

Multiple Choice (2)

What will be the output when the following code is evaluated?

```
for (int k=0; k<3; k++){
    for (int j=1; j<4; j++){
        System.out.print(j+” “);
    }
    System.out.println();
}
```

Multiple Choice (2)

Ⓐ 1 2 3 4
1 2 3 4
1 2 3 4

Ⓑ 0 1 2
0 1 2
0 1 2

Ⓒ 1 2 3
1 2 3
1 2 3

Ⓓ 1 2 3
1 2 3
1 2 3
1 2 3

Multiple Choice (2)

Ⓐ 1 2 3 4
1 2 3 4
1 2 3 4

Ⓑ 0 1 2
0 1 2
0 1 2

Ⓒ 1 2 3
1 2 3
1 2 3

Ⓓ 1 2 3
1 2 3
1 2 3
1 2 3

Multiple Choice (3)

Consider the following code segments. Assume `someNum` has been correctly defined and initialized as a positive integer.

```
I. int i;
   for (i=0; i<someNum; i++){
     someNum--;
   }
II. int i;
   for (i=1; i<someNum-1; i++){
     someNum-=1;
   }
III. int i=0;
    while (i<someNum){
     i++;
     someNum--;
   }
```

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Multiple Choice (3)

Ⓐ The for loops in I and II do not have the same number of iterations.

Ⓑ The for loop in I and the while loop in III have the same number of iterations.

Ⓒ The value of i after the execution of II and III is the same.

Ⓓ I, II, and III produce identical results in `someNum`.

Multiple Choice (3)

Ⓐ The for loops in I and II do not have the same number of iterations.

Ⓑ The for loop in I and the while loop in III have the same number of iterations.

Ⓒ The value of i after the execution of II and III is the same.

Ⓓ I, II, and III produce identical results in `someNum`.

Multiple Choice (4)

Consider the following code segments:

```
for (int i=1; i<100; i = i*2){
    if (i/50 ==0)
        System.out.print(i + " ");
}
```

What is printed as result of executing the code segment?

Multiple Choice (4)

Ⓐ 1 2 4 8 16 32 64

Ⓑ 1 2 4 8 16 32

Ⓒ 2 4 8 16 32 64

Ⓓ 2 4 8 16 32

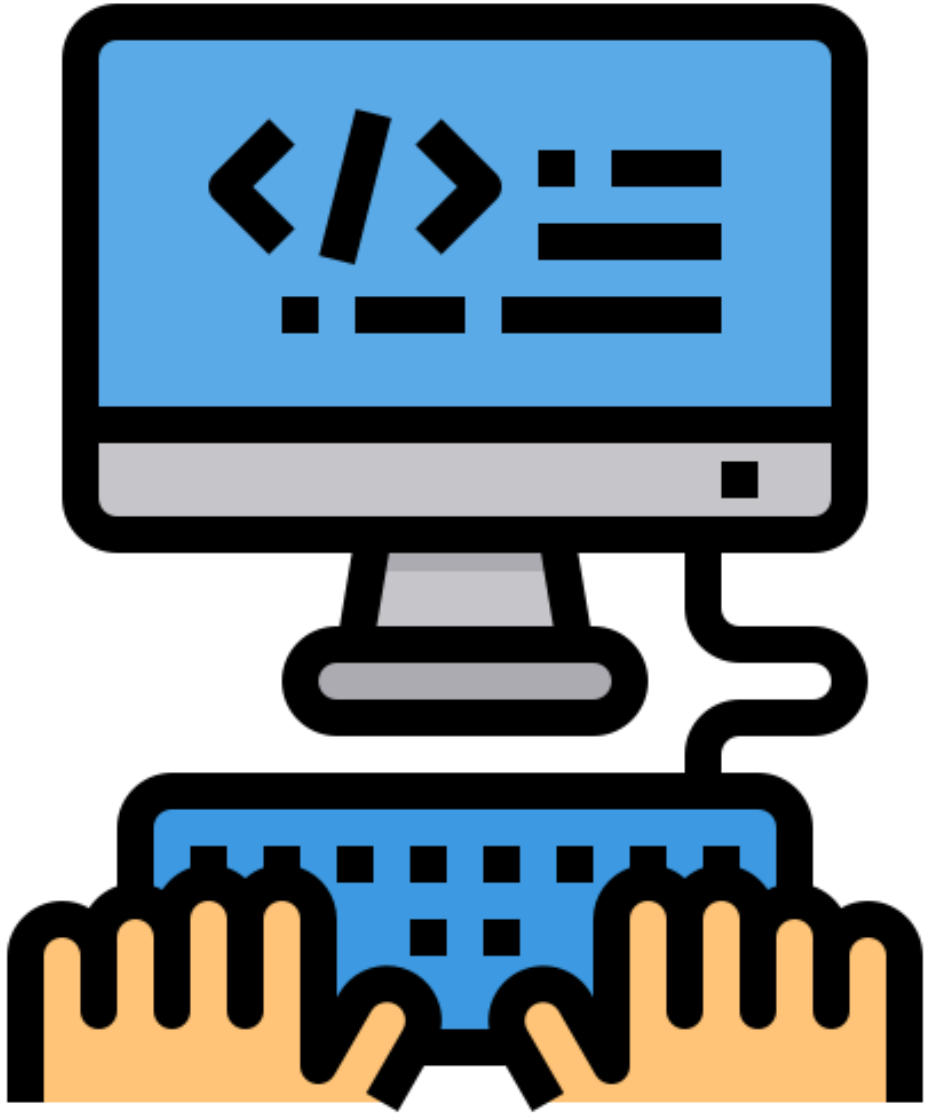
Multiple Choice (4)

Ⓐ 1 2 4 8 16 32 64

Ⓑ 1 2 4 8 16 32

Ⓒ 2 4 8 16 32 64

Ⓓ 2 4 8 16 32



For-each Statement

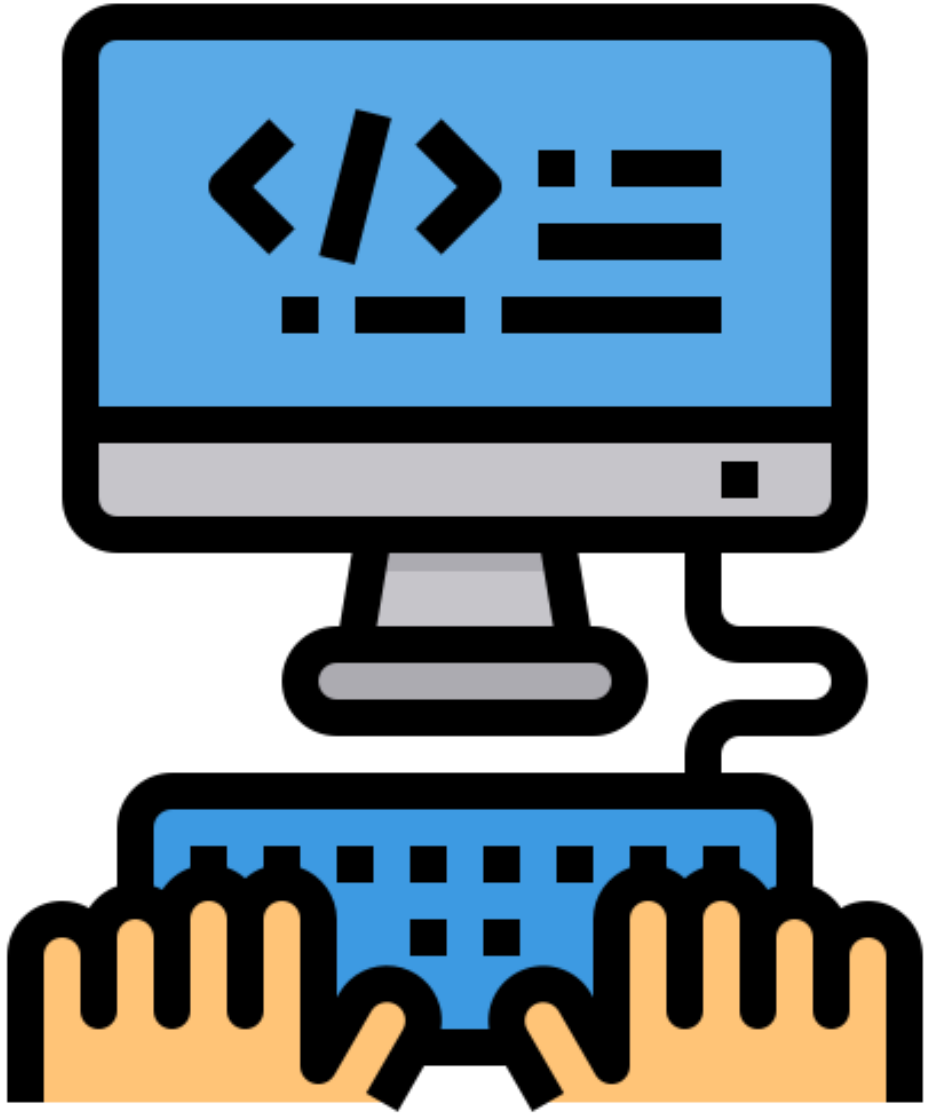
SECTION 4

For-Each Statement

```
for (datatype x: data_collections){
    // do something.
}
```

Iterable Data Collections

- No removal or deletion in for-each loop. **ConcurrentModificationException** will be raised.
- Data must be iterable



Practice MCQ Questions

SECTION 5

Multiple Choice (1)

Consider the following output:

```
0 1
0 2 4
0 3 6 9
0 4 8 12 16
```

Which of the following code segment will produce this output?

Multiple Choice (1)

Ⓐ
```
for (int x=1; x<5; x++){
    for (int z=0; z<=x; z++){
      System.out.print(x*z+" ");
    }
    System.out.println(" ");
}
```

Ⓑ
```
for (int x=1; x<=5; x++){
    for (int z=0; z<x; z++){
      System.out.print(x*z+" ");
    }
    System.out.println(" ");
}
```

Ⓒ
```
for (int x=1; x<5; x++){
    for (int z=0; z<=4; z++){
      System.out.print(x*z+" ");
    }
    System.out.println(" ");
}
```

Ⓓ
```
for (int x=1; x<=5; x++){
    for (int z=0; z<=x; z++){
      System.out.print(x*z+" ");
    }
    System.out.println(" ");
}
```

Multiple Choice (1)

Ⓐ
```
for (int x=1; x<5; x++){
    for (int z=0; z<=x; z++){
      System.out.print(x*z+" ");
    }
    System.out.println(" ");
}
```

Ⓑ
```
for (int x=1; x<=5; x++){
    for (int z=0; z<x; z++){
      System.out.print(x*z+" ");
    }
    System.out.println(" ");
}
```

Ⓒ
```
for (int x=1; x<5; x++){
    for (int z=0; z<=4; z++){
      System.out.print(x*z+" ");
    }
    System.out.println(" ");
}
```

Ⓓ
```
for (int x=1; x<=5; x++){
    for (int z=0; z<=x; z++){
      System.out.print(x*z+" ");
    }
    System.out.println(" ");
}
```

Multiple Choice (2)

The speed limit of a stretch of highway is 55 miles per hour (mph). The highway patrol issues speeding tickets to anyone caught going faster than 55 miles per hour. The fine for speeding is based on the following scale:

Speed	**Fine**
greater than 55 mph but less than 65 mph	$100
greater than or equal to 65 mph but less than 75 mph	$150
greater than or equal to 75 mph	$300

Multiple Choice (2)

If the value of `int` variable `speed` is the speed of a driver who was pulled over for going faster than 55 mph, which of the following code segments will assign the correct value to the int variable `fine`?

```
I. if (speed >=75)fine = 300;
   if (speed >=65 && speed < 75)fine = 150;
   if (speed > 55 && speed <65) fine = 100;

II. if (speed >=75)fine = 300;
    if (65 <= speed < 75)fine = 150;
    if (55 < speed < 65) fine = 100;
III. if (speed >= 75) fine = 300;
    if (speed >= 65) fine = 150;
    if (speed > 55) fine = 100;
```

Multiple Choice (2)

Ⓐ I only

Ⓑ Ⅲ only

Ⓒ I and Ⅱ

Ⓓ I and Ⅲ

Multiple Choice (2)

Ⓐ I only

Ⓑ Ⅲ only

Ⓒ I and Ⅱ

Ⓓ I and Ⅲ

Multiple Choice (3)

Consider the following code segment:

```
int x = 10;
int y = 3;
boolean b = true;
for (int i=0; i<15; i += 5){
    x = x + y;
    b = (x % y == 2);
    if (!b) {
        y++;
        i += 5;
    }
}
```

Multiple Choice (3)

Which of the value of `x` after the code segment executes?

Ⓐ 10

Ⓑ 15

Ⓒ 17

Ⓓ 22

Multiple Choice (3)

Which of the value of `x` after the code segment executes?

Ⓐ 10

Ⓑ 15

Ⓒ 17

Ⓓ 22

Multiple Choice (4)

In the following statement, `a` and `b` are boolean variables:

```
Boolean c = (a && b) || !(a || b);
```

Under what conditions will the value of c be true?

Multiple Choice (4)

Ⓐ Only when the value of `a` is different than the value of `b`.

Ⓑ Only when the value of `a` is the same as the value of `b`.

Ⓒ Only when `a` and `b` are both `true`.

Ⓓ The value of `c` will be `true` for all values of `a` and `b`.

Multiple Choice (4)

Ⓐ Only when the value of `a` is different than the value of `b`.

Ⓑ Only when the value of `a` is the same as the value of `b`.

Ⓒ Only when `a` and `b` are both `true`.

Ⓓ The value of `c` will be `true` for all values of `a` and `b`.

Multiple Choice (5)

Consider the following code segment:

```
while ((x > y) || y >= z){
  // code not shown
}
```

In the code segment above, `x`, `y`, and `z` are variables of type `int`. Which of the following must be true after the code segment executed?

Multiple Choice (5)

Ⓐ `x > y || y >= z`

Ⓑ `x <= y || y > z`

Ⓒ `x < y && y <= z`

Ⓓ `x <= y && y < z`

Multiple Choice (5)

Ⓐ `x > y || y >= z`

Ⓑ `x <= y || y > z`

Ⓒ `x < y && y <= z`

Ⓓ `x <= y && y < z`

Multiple Choice (6)

Consider the following code segment:

```
int a = 0;
for (int i=0; i<10; i++){
    for (int k=0; k<=5; k++){
        for (int z=1; z<=16; z = z*2){
            a++;
        }
    }
}
```

Which is the value of `a` after the code segment executes?

Multiple Choice (6)

Ⓐ 31

Ⓑ 180

Ⓒ 200

Ⓓ 300

Multiple Choice (6)

Ⓐ 31

Ⓑ 180

Ⓒ 200

Ⓓ 300

Multiple Choice (7)

Consider the following code segment:

```
int x=10;
int y = x / 3;
int z = x % 2;
x++;
System.out.println(x);
```

What is printed as a result of executing the code segment above?

Multiple Choice (7)

Ⓐ 2

Ⓑ 10

Ⓒ 11

Ⓓ 15

Multiple Choice (7)

Ⓐ 2

Ⓑ 10

Ⓒ 11

Ⓓ 15

Multiple Choice (8)

Consider the following code segment:

```
int a=10;
double b = 3.7;
int c = 4;
int x = (int)(a+b);
double y = (double) a/c;
double z = (double)(a/c);
double w = x + y + z;
System.out.println(w);
```

What is printed as a result of executing the code segment above?

Multiple Choice (8)

Ⓐ 10

Ⓑ 15.5

Ⓒ 17

Ⓓ 17.5

Multiple Choice (8)

Ⓐ 10

Ⓑ 15.5

Ⓒ 17

Ⓓ 17.5

Multiple Choice (9)

Consider the following code segment:

I.
```
int x = 10;
int y = 20;
int z = 0;
if (x < y && 10 < y /z)
 System.out.println("Homer");
else{
 System.out.println("Bart");
```

II.
```
int x = 10;
int y = 20;
int z = 0;
if (x > y && 10 < y /z)
 System.out.println("Homer");
else{
 System.out.println("Bart");
```

III.
```
int x = 10;
int y = 20;
int z = 0;
if (x < y || 10 < y /z)
 System.out.println("Homer");
else{
 System.out.println("Bart");
```

Multiple Choice (9)

Which of the code segments above will run without error?

Ⓐ I only

Ⓑ Ⅱ only

Ⓒ Ⅲ only

Ⓓ Ⅱ and Ⅲ

Multiple Choice (9)

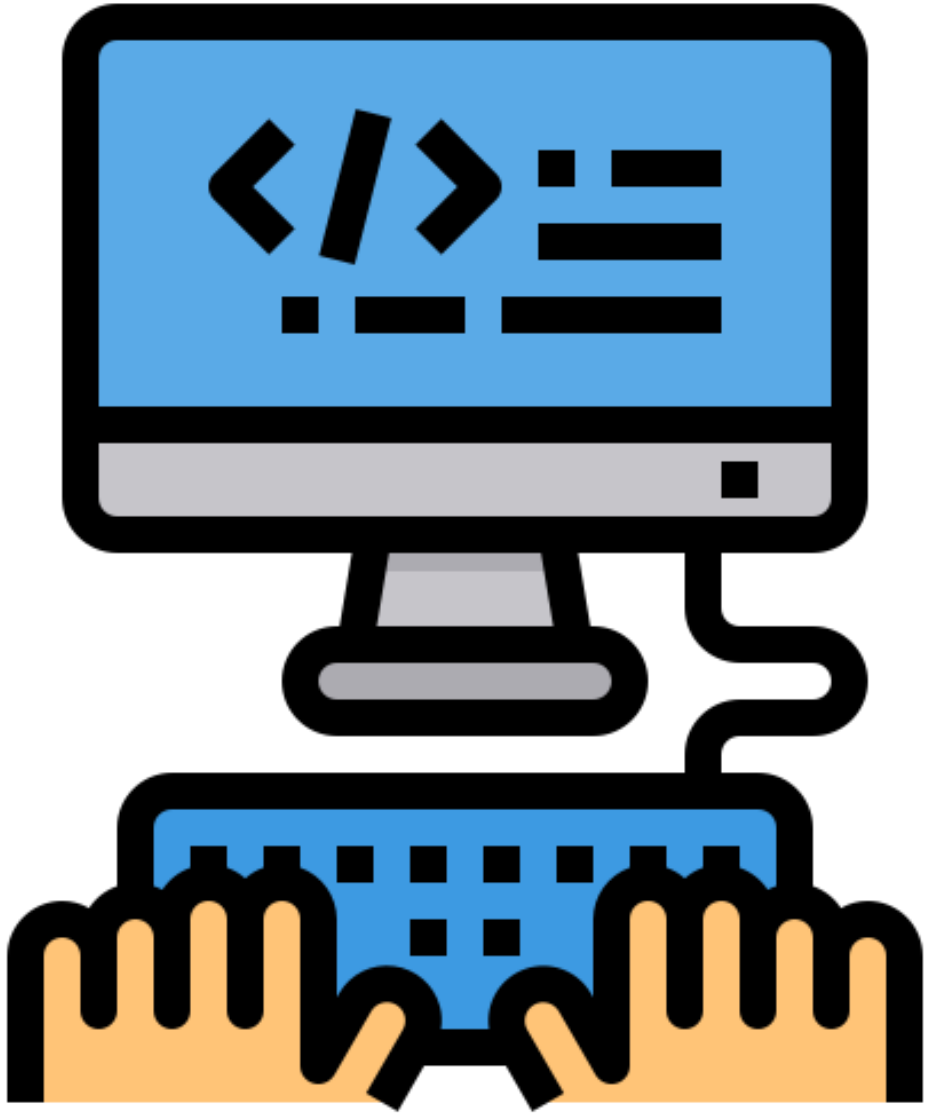
Which of the code segments above will run without error?

Ⓐ I only

Ⓑ II only

Ⓒ III only

Ⓓ II and III



Loop Algorithms

SECTION 6

Index Calculation: First N, Last N

```
String s1 = str.substring(0, n);
String sf = str.substring(str.length()-n);
```

Index Calculation: Middle Index

Middle Left:

```
int low = 0, high = str.length()-1;
int mid = (low + high)/2;
```

Middle Right:

```
int mid = str.length()/2;
```

Linear Search

```
public int search(int[] a, int key){
    for (int i=0; i<a.length; i++){
        if (a[i] == key) return I;
    }
   return -1;
}
```

Binary Search

```
public int search(int[] a, int key){
    int low = 0;
    int high = a.length-1;
    while (low <= high){
      int mid = (low+high)/2;
      if (a[mid]==key) return mid;
      else if (a[mid]>key) high = mid-1;
      else low = mid+1;
    }
    return -1-low;
}
```

Sum

```
public int search(int[] a){
    int s=0;
    for (int i=0; i<a.length; i++){
      s += a[i];
    }
    return s;
}
```

Max

```
public int search(int[] a){
    int m=Integer.MIN_VALUE;
    for (int i=0; i<a.length; i++){
      if (a[i]>m) m = a[i];
    }
    return m;
}
```

Max Index

```
public int search(int[] a){
    int idx=-1;
    for (int i=0; i<a.length; i++){
      if (a[i]>a[idx]) idx=i;
    }
    return idx;
}
```

Sentinel Loop

```
Scanner input = new Scanner(System.in);
int s = 0;
String line = input.nextLine();
while(!line.toLowerCase().equals("exit")){
     int x = Integer.parseInt(line);
     s += x;
     line = input.nextLine();
}
return s;
```

Print a List

```
public String toString(int[] a){
    String s = "<";
    for (int i=0; i<a.length; i++){
     if (i==0) s += a[i];
    else s += ", "+ a[i];
   }
   s += ">";
   return s;
}
```

Leap Frog

```
public int steps(int limit){
     int count;
     int total=0;
     while (total < limit){
         int step = (int)(Math.raondom()*5);  // 0 - 4 per step
         total += step;
        count++;
    }
    return count;
}
```

Top 3

```
public int[] pickTo3(int[] a){ // a.length >=3, no duplicates
   int gold = Integer.MIN_VALUE;
   int silver = Integer.MIN_VALUE;
   int bronze = Integer.MIN_VALUE;
   for (int i=0; i<a.length; i++){
     if (a[i]>gold){
        bronze = silver;
        silver = gold;
        gold = a[i];
     } else if (a[i]>silver){
        bronze = silver;
        silver = a[i];
     } else if (a[i]>bronze){
        bronze = a[i];
     }
   }
   return new int[]{gold, silver, bronze};
}
```